



**CREATIVE  
GAMING**

**GROUP 15 PRESENTATION**

**UPLIFT**

**MODELING**

**Emma Wu**

**Yoonjeong Park**

**Kuan-Ling Tseng**

**Yijiang Qian**

# Project Overview

## Introduction to the marketing agency

- After the success of the work on propensity-to-buy modeling, our team is going to focus on uplift modeling to pick customers to target for the Zalon campaign.

## Why Uplift Modeling?

- Uplift modeling is crucial in targeted marketing because it allows marketers to identify the specific customers who are most likely to positively respond to a marketing treatment

## Objectives

- Use various machine learning models for uplift predictions
- Identify customer to target
- Determine the optimal proportion
- Compare with Propensity-to-Buy models

# **PART I**

## **Uplift Modeling Using Machine Learning**

# Data Preparation

## Two Data Sets Used

- cg\_organic\_control  
(30,000 customers, no ad)
- cg\_ad\_random  
(30,000 customers, received ad)  
(treatment data)

```
pd.read_parquet("data/cg_organic_control.parquet")
```

```
pd.read_parquet("data/cg_ad_random.parquet")
```

## Add New Variable

- 'ad' (1 for treated, 0 for control)

```
cg_ad_random['ad'] = 1
```

```
cg_organic_control['ad'] = 0
```

## Modify Datasets

- Combined both datasets into  
cg\_rct\_stacked
- Training (70%) vs. Testing (30%)  
split using converted & ad

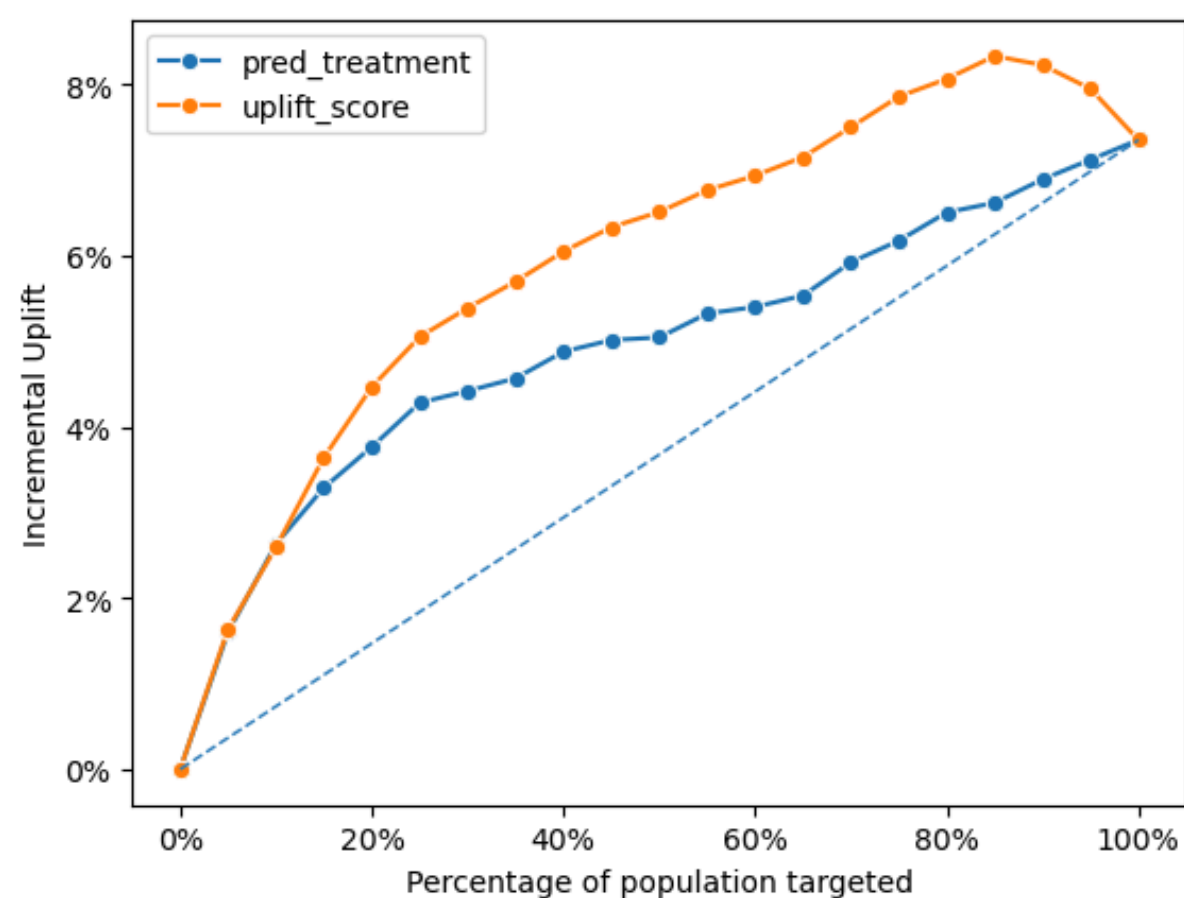
```
pd.concat([cg_organic_control, cg_ad_random])
```

```
test_size= 0.3, random_state= 1234,
```

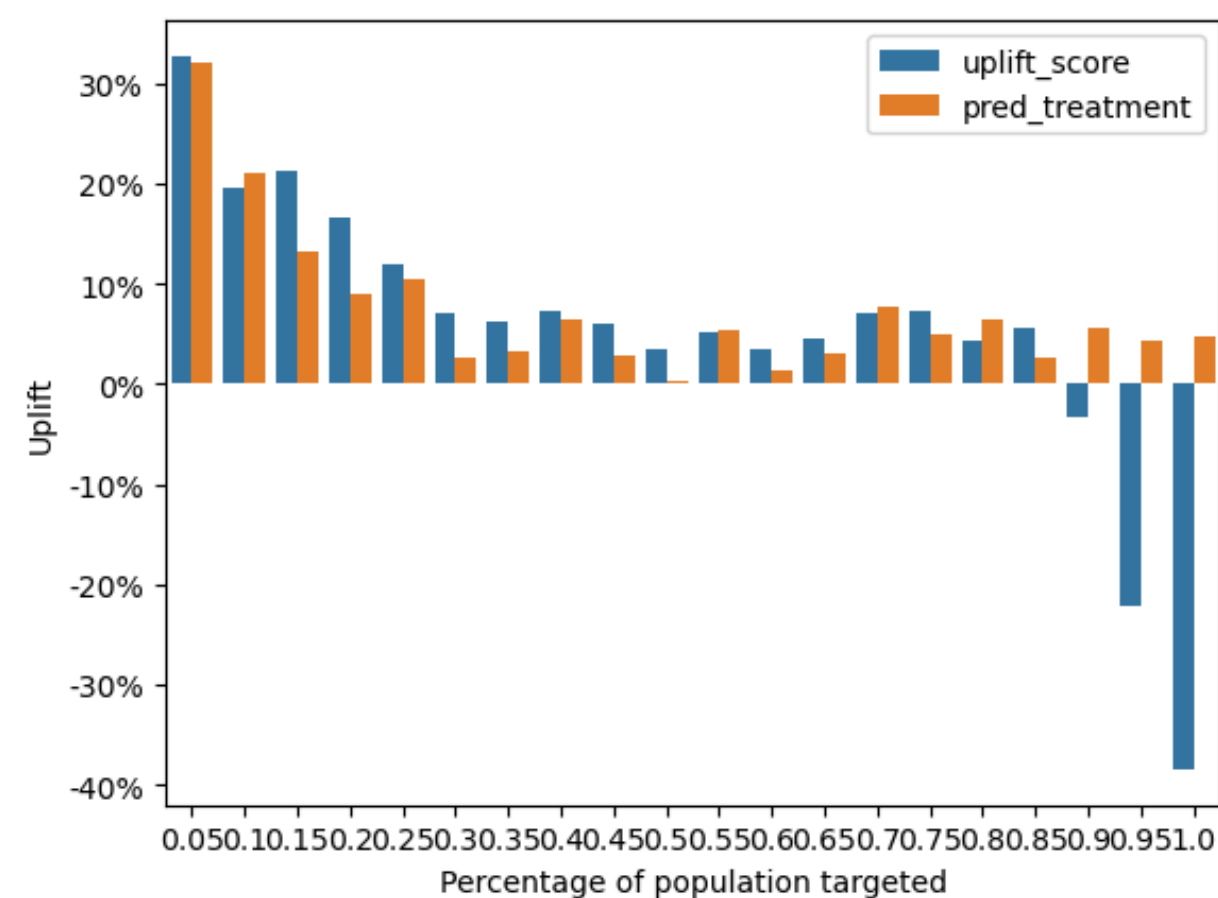
```
strat_var= ['converted', 'ad']
```

# Using Logistic Regression

## Incremental Uplift (%)



## Uplift (%)



## Incremental Profit

Incremental profit: 45907.759

Propensity incremental profit: 32065.482

difference = inc\_profit\_uplift - inc\_profit\_propensity

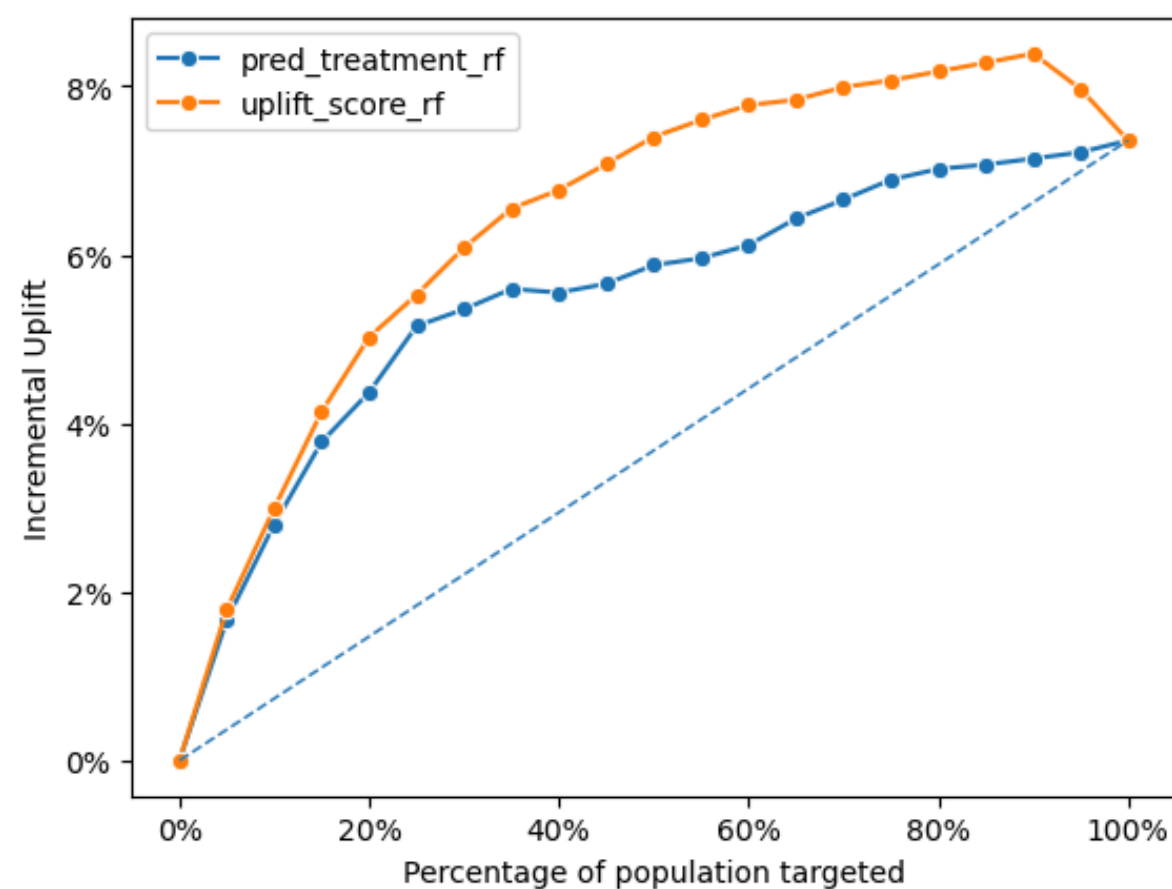
13842.276

```
cg_rct_stacked['uplift_score'] = cg_rct_stacked.pred_treatment - cg_rct_stacked.pred_control
```

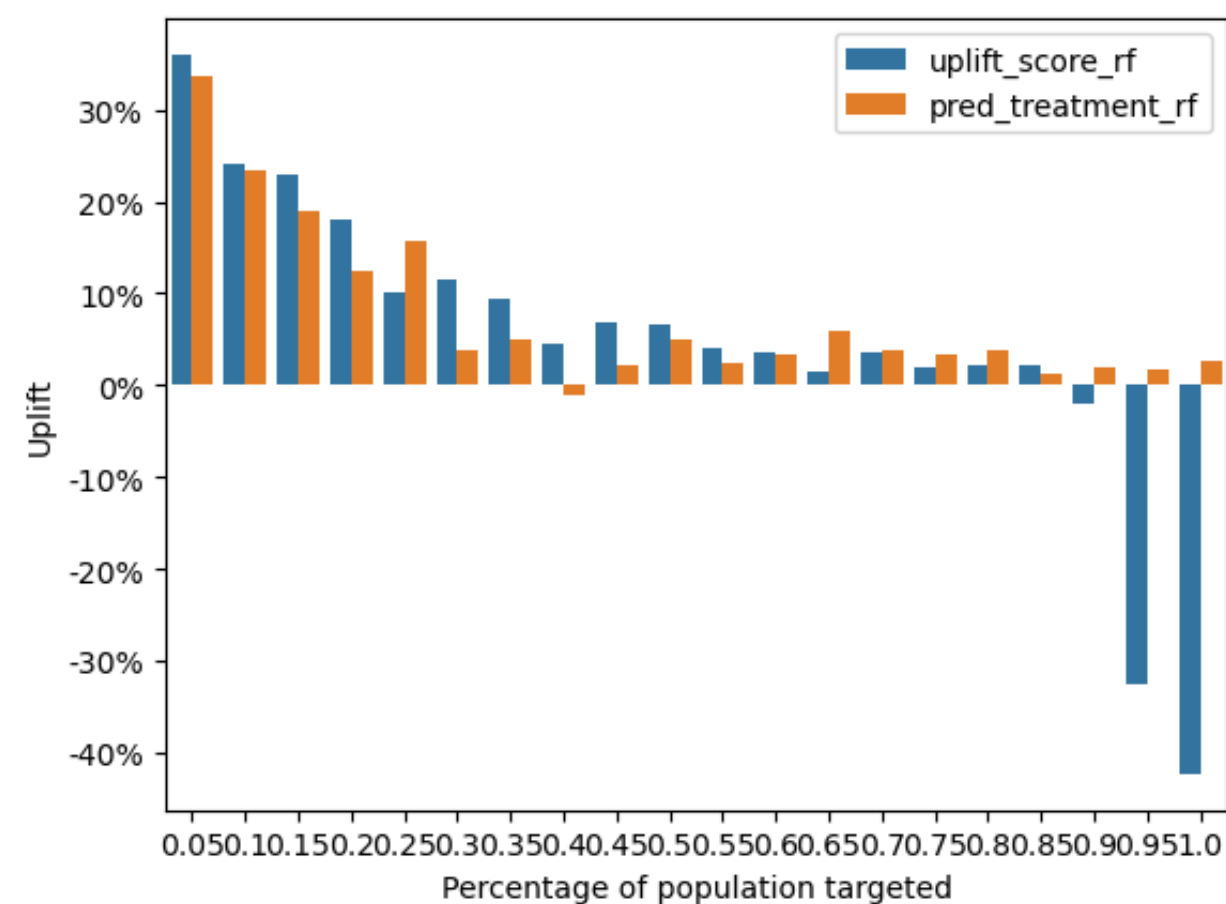


# Using Random Forest

## Incremental Uplift (%)



## Uplift (%)



## Incremental Profit

Incremental profit: 54374.147

Propensity Incremental profit: 47697.137

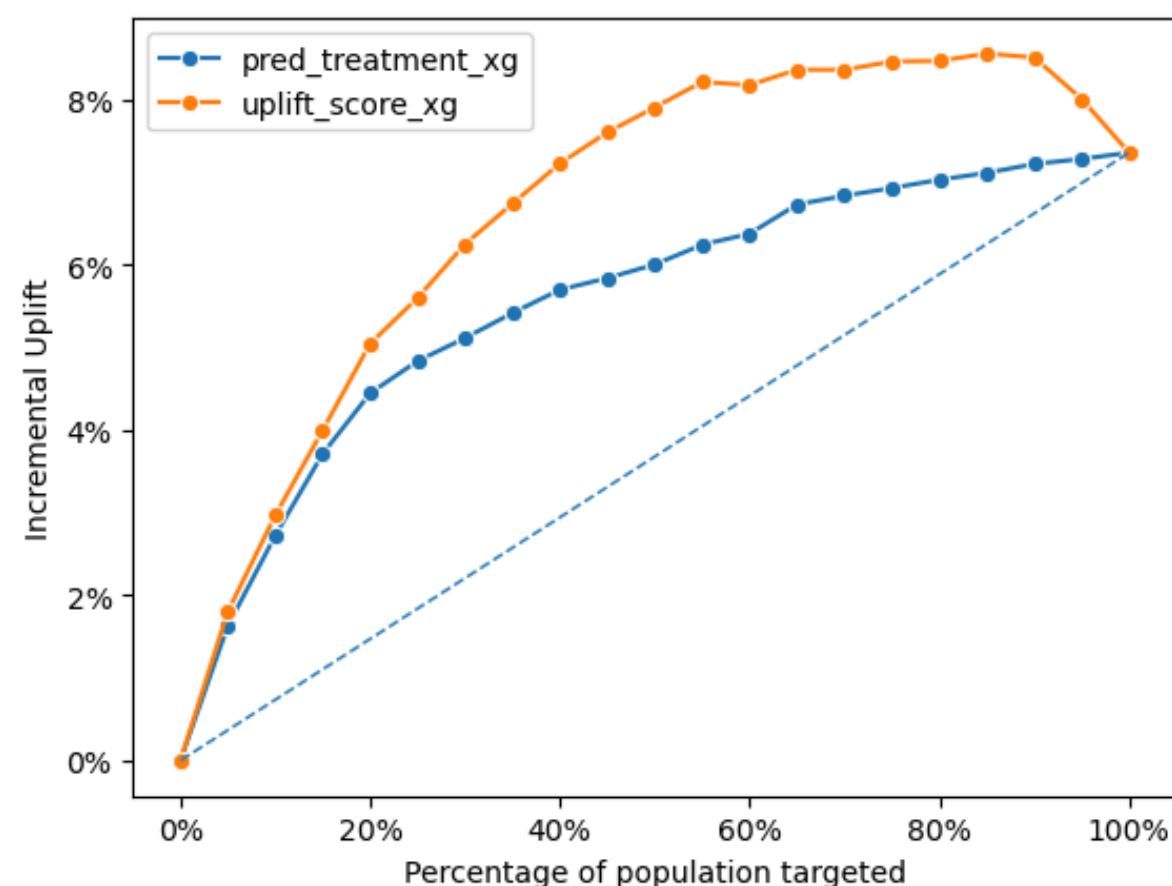
difference = inc\_profit\_uplift - inc\_profit\_propensity

6677.010

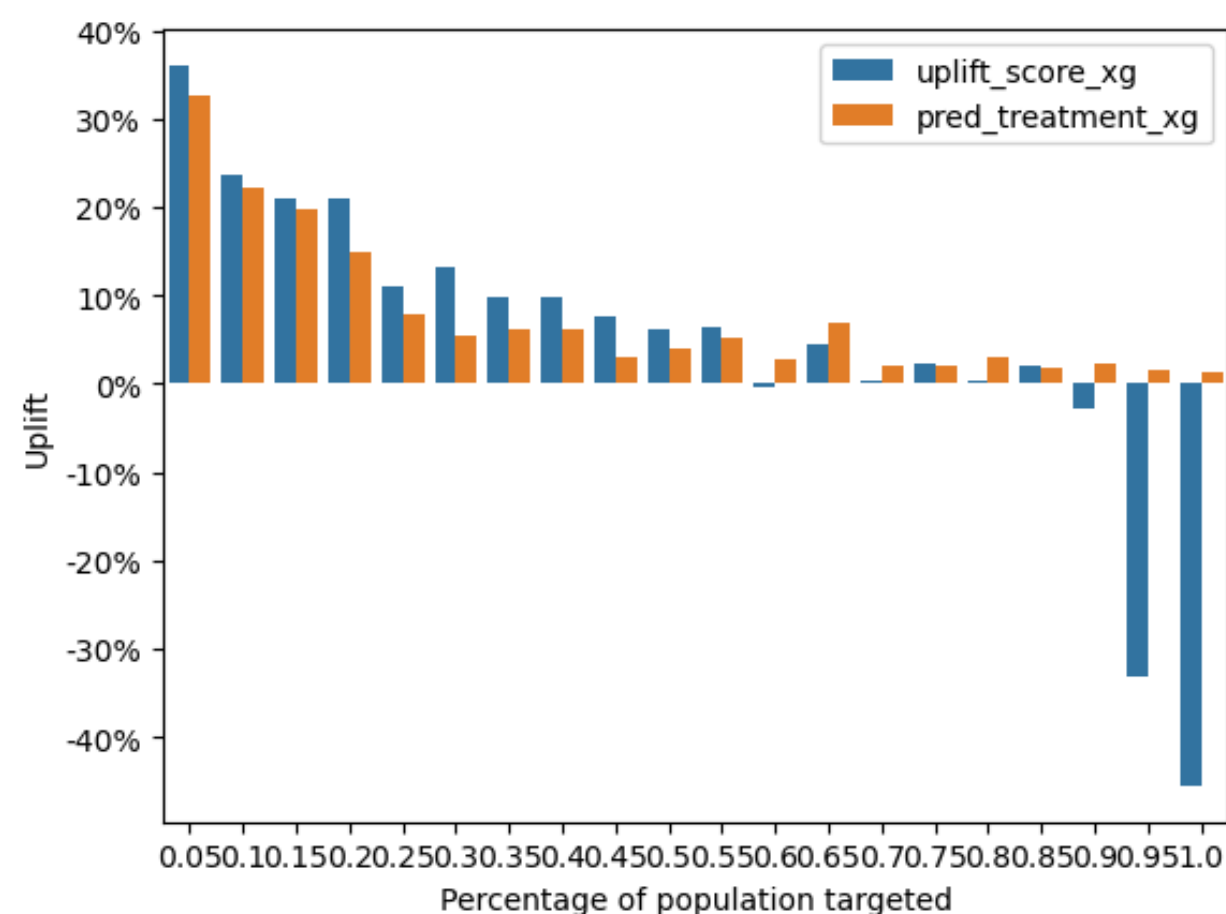
```
cg_rct_stacked['uplift_score_rf'] = cg_rct_stacked.pred_treatment_rf - cg_rct_stacked.pred_control_rf
```

# Using XGBoost Model

## Incremental Uplift (%)



## Uplift (%)



## Incremental Profit

Incremental profit: 55694.019

Propensity Incremental profit: 41965.615

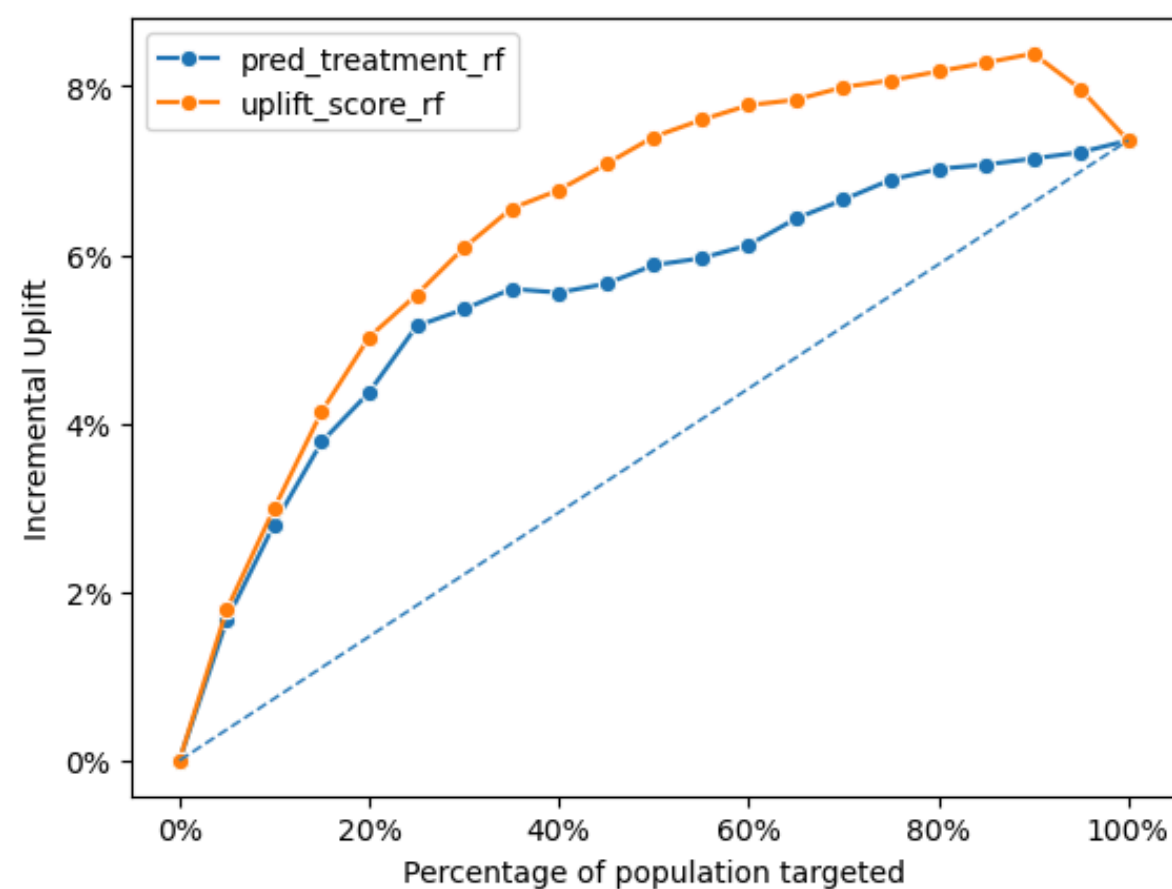
difference = inc\_profit\_uplift - inc\_profit\_propensity

13728.404

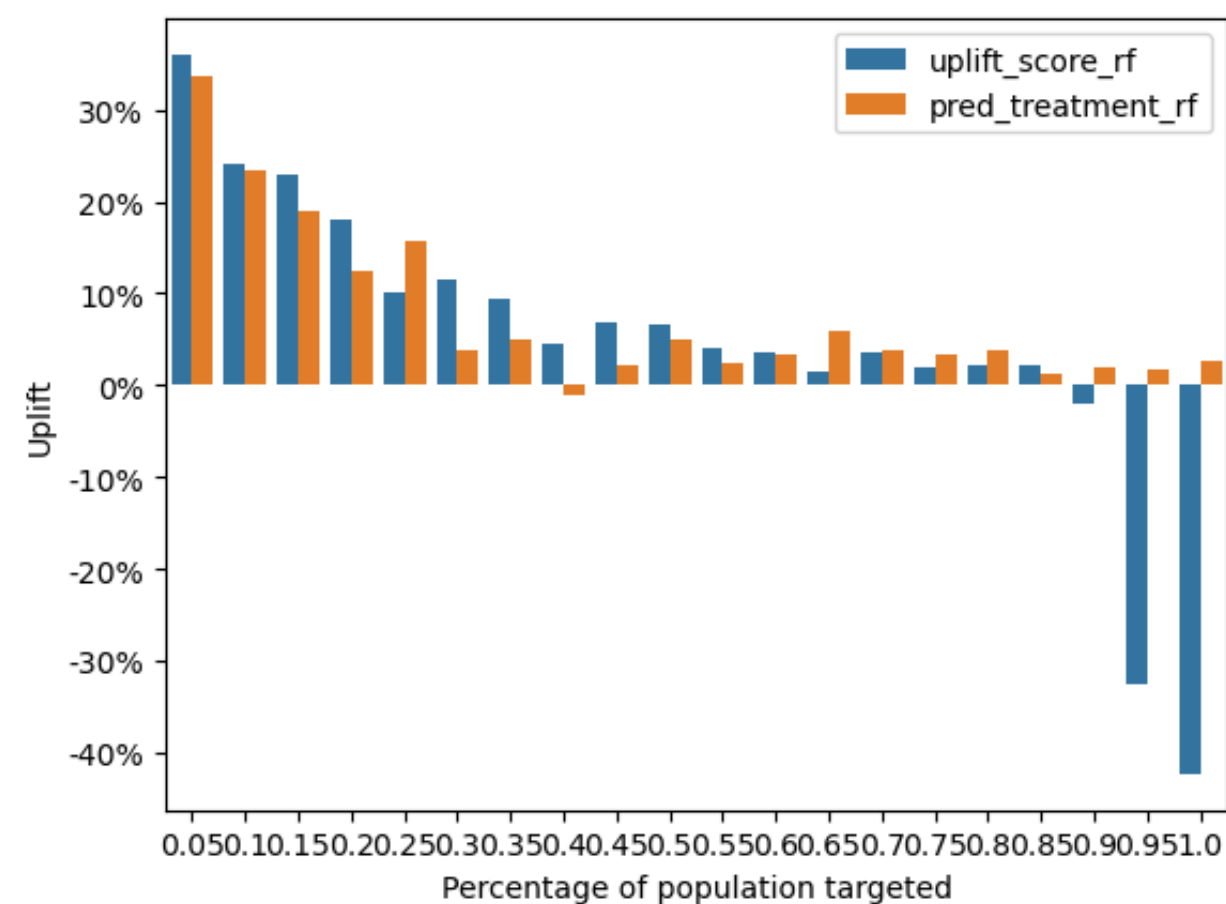
```
cg_rct_stacked['uplift_score_xg'] = cg_rct_stacked.pred_treatment_xg - cg_rct_stacked.pred_control_xg
```

# Using Random Forest

## Incremental Uplift (%)



## Uplift (%)



## Incremental Profit

Incremental profit: 54374.147

Propensity Incremental profit: 47697.137

difference = inc\_profit\_uplift - inc\_profit\_propensity

6677.010

```
cg_rct_stacked['uplift_score_rf'] = cg_rct_stacked.pred_treatment_rf - cg_rct_stacked.pred_control_rf
```



# Uplift VS Propensity-to-Buy Model

Overall, Uplift model consistently achieves a higher uplift and incremental uplift than the propensity model. The reason is that when we target the highest probability converters, we also include many people who have converted in a control, so the incremental gain is lower. On the other hand, the uplift model specifically includes those who are truly persuadable, leading to higher incremental conversion in the top bins.

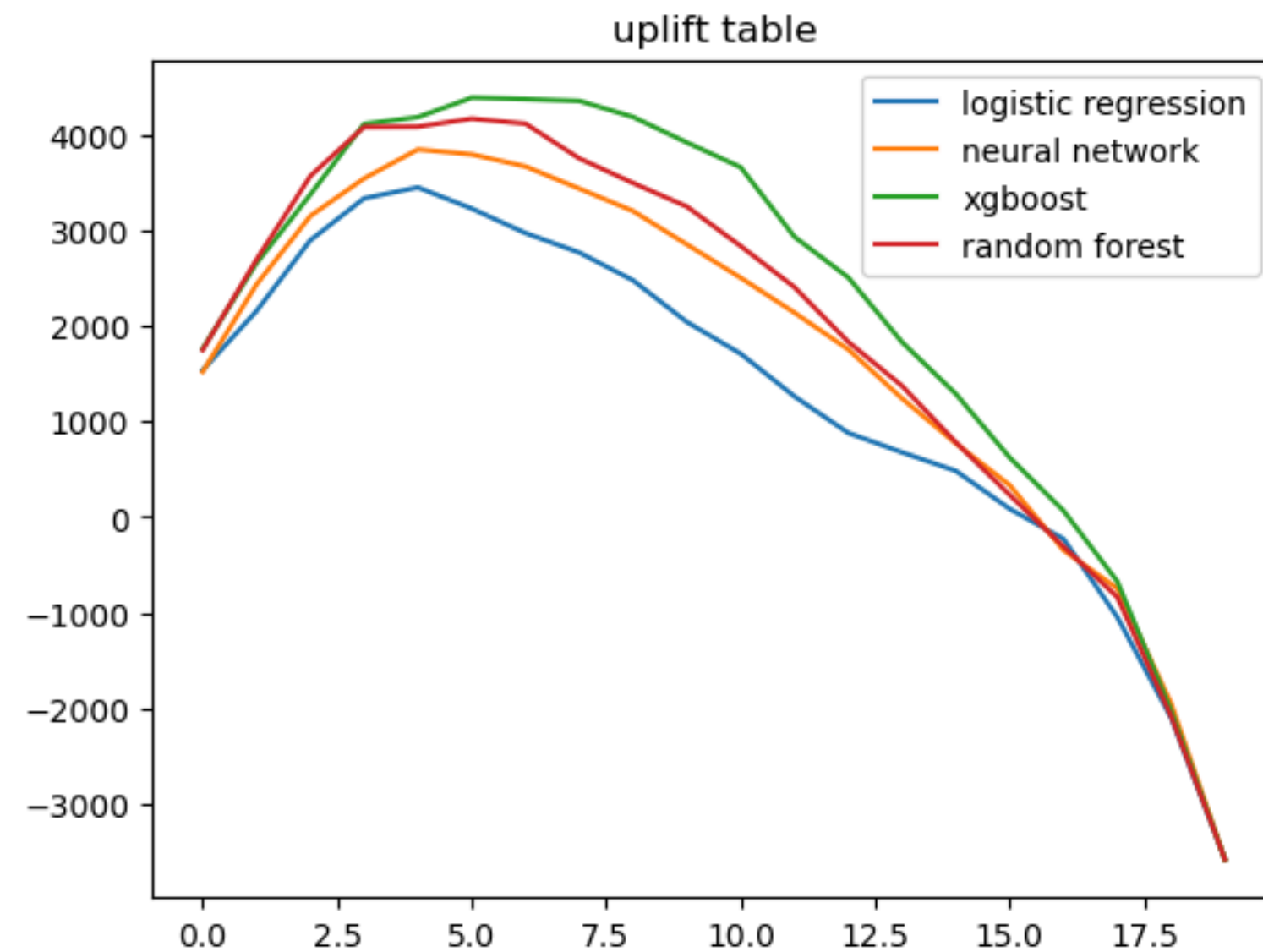
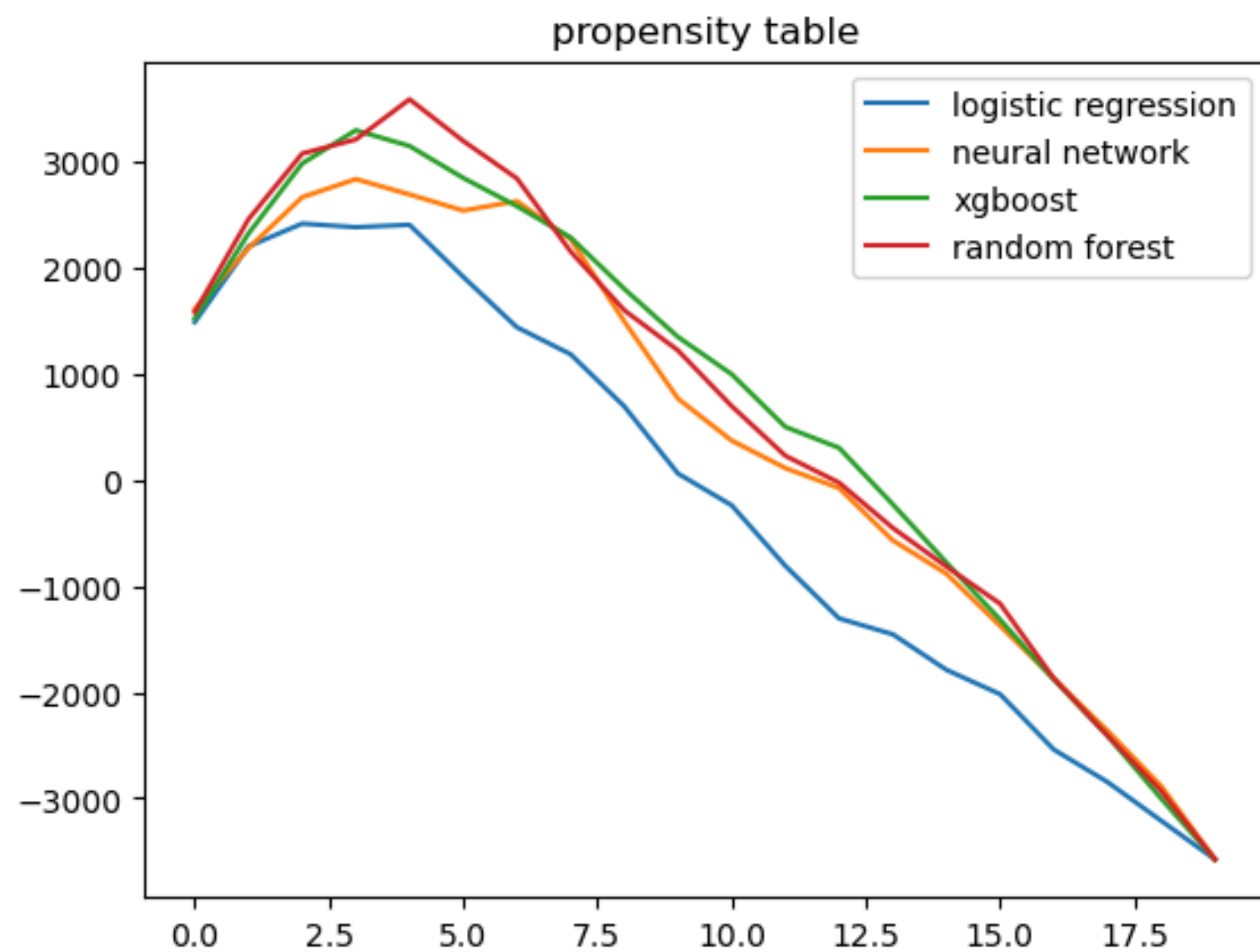
**Uplift captures incremental response**

**Propensity model focuses on likelihood of conversion**

## **PART II**

# **Targeting the Optimal Proportion of Customers**

# Uplift VS Propensity-to-Buy Model



# Uplift VS Propensity-to-Buy Model

| Model               | Propensity Optimal Proportion | Uplift Optimal Proportion | Propensity Optimal Profit | Uplift Optimal Profit |
|---------------------|-------------------------------|---------------------------|---------------------------|-----------------------|
| Logistic Regression | 15%                           | 25%                       | 32212.83                  | 45907.76              |
| Neural Network      | 20%                           | 25%                       | 37807.77                  | 51194.43              |
| Random Forest       | 25%                           | 30%                       | 47837.13                  | 55483.28              |
| XGBoost             | 20%                           | 30%                       | 43947.86                  | 58395.95              |

- The optimal targeting proportion is higher for uplift models (25-30%), meaning more customers can be profitably reached compared to propensity models (15-25%).
- Uplift models result in a higher profit at an optimal proportion of targeting customer. With maximum 58395.95.

# Conclusion

## Uplift Modeling Identifies True Causal Impact of Advertising

- Propensity models predict likelihood of purchase, but they do not differentiate between customers who would buy anyway and those influenced by the ad.
- Uplift models explicitly capture the difference between treated (ad-exposed) and control groups, ensuring that only customers who are positively influenced by the ad are targeted.



## Uplift Models Optimize Targeting Strategy for Maximum Revenue

- The table shows that across all models, uplift modeling results in higher profits at an optimal proportion of targeted customers
- The optimal targeting proportion is also higher for uplift models (25-30%), meaning more customers can be profitably reached compared to propensity models (15-25%).





# Conclusion

- Uplift modeling optimally selects customers who are truly influenced by advertising, while propensity models waste resources on natural buyers. This leads to higher incremental revenue and better cost efficiency.





THANK  
YOU